192425037

EXP14:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#include <stdlib.h>

/* read a line and keep only letters, converted to lower case */
int readletters(char *out) {
    char line[200];
    int i, n = 0;
    fgets(line, sizeof line, stdin);
    for (i = 0; line[i]; i++) if (isalpha(line[i])) out[n++] = tolower(line[i]);
    out[n] = '\0';
    return n;
}

int main() {
    char pt[100], ct[100], target[100];
    int key[100], kn = 0, n, i, tn;
    char buf[300], *tok;

    /* (a) encrypt with the given key stream */
    printf("Enter plaintext: ");
    n = readletters(pt);
    printf("Enter key stream (numbers): ");
    fgets(buf, sizeof buf, stdin);
    for (tok = strtok(buf, " \n"); tok; tok = strtok(NULL, " \n")) key[kn++] = atoi(tok);

    for (i = 0; i < n; i++) ct[i] = (pt[i] - 'a' + key[i]) % 26 + 'A';
```

```
    ct[n] = '\0';
    printf("Ciphertext: %s\n\n", ct);

    /* (b) find a key so the same ciphertext decrypts to another plaintext */
    printf("Enter desired plaintext: ");
    tn = readletters(target);
    printf("Key stream required: ");
    for (i = 0; i < tn; i++)
        printf("%d ", ((ct[i] - 'A') - (target[i] - 'a') + 26) % 26);
    printf("\nDecrypted with this key: ");
    for (i = 0; i < tn; i++) {
        int k = ((ct[i] - 'A') - (target[i] - 'a') + 26) % 26;
        putchar((ct[i] - 'A' - k + 26) % 26 + 'a');
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
Enter plaintext: send more money
Enter key stream (numbers): 9 0 1 7 23 15 21 14 11 11 2 8 9
Ciphertext: BEOKJDMSXZPMH

Enter desired plaintext: cash not needed
Key stream required: 25 4 22 3 22 15 19 5 19 21 12 8 4
Decrypted with this key: cashnotneeded
```